# Basin Geometry and Autonomous Viability in a Viable System Kernel

Empirical Characterization of Attractor Structure in vOS

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**Abstract**

## 1 Introduction

### 1.1 Background: Viable System Model and vOS

## 2 Experimental Setup

### 2.1 Hardware

Table 1: Experimental hardware and software environment.

| Component | Specification |
|---|---|
| CPU | Intel Core i7-7600U @ 2.80 GHz, 2 cores / 4 threads |
| RAM | 24 GB DDR4-2133 |
| Platform | Proxmox VE 8 (KVM hypervisor, bare-metal) |
| OS | Debian 12 (Bookworm) VM |
| Cgroup | cgroup v2 (unified hierarchy) |
| Binary | `vsf-nucleus` v0.0.1 (Rust, release build) |

### 2.2 Kernel Parameters

$$\text{viable}(V,t) \;=\; \cos(\boldsymbol{x}(t),\, \boldsymbol{s}) \;\geq\; 0.70 \;\wedge\; H(t) \;\leq\; H_{\max} \tag{1}$$

Table 2: Experiments conducted. All runs on 2026-04-13, n02 (Proxmox).

| ID | Name | Question | Ticks |
|---|---|---|---|
| EXP-2 | `multikernel_geometry` | Basin uniqueness (Q1) | 1,000 |
| EXP-3 | `s3_failure_resilience` | Autonomous viability (Q2) | ∼500 |
| EXP-G | `critical_angle` | PSI threshold bound (Q3) | ∼800 |
| EXP-1 | `spec_boot` 10k | Long-run compression | 10,000 |

Table 3: Per-kernel results after 1,000 ticks (`multikernel_geometry`).

| **Kernel** | $\tau_0$ | $\tau_f$ | $\cos(\boldsymbol{x}, \boldsymbol{s})$ | **Soplos** | $\overline{\Delta L}$ |
|---|---|---|---|---|---|
| $k_0$ | 0.55 | 0.038 | 0.8701 | 14 | −0.00915 |
| $k_1$ | 0.65 | 0.141 | 0.8822 | 14 | −0.01053 |
| $k_2$ | 0.75 | 0.246 | 0.8959 | 14 | −0.01199 |
| $k_3$ | 0.85 | 0.352 | 0.9096 | 14 | −0.01342 |

### 2.3 Experiment Overview

## 3 Basin Geometry: Single Attractor Under Varied Initial Conditions

### 3.1 Results

### 3.2 Discussion

## 4 Autonomous Viability Without S3 Coordination

### 4.1 Results

### 4.2 Discussion

## 5 Bounding the PSI Intervention Threshold

### 5.1 Results

### 5.2 Discussion

$$\mathrm{PSI}(V,t) = \mathbf{1}[\,\cos(\boldsymbol{x}(t),\, \boldsymbol{s}) < \theta_{\mathrm{PSI}} \;\wedge\; \mathrm{viable}(V,t) = \mathrm{false}\,] \tag{2}$$

## 6 Preliminary Observations on Long-Run Dynamics

## 7 Discussion

### 7.1 Future Work

## 8 Conclusion

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Table 4: Pairwise cosine similarity between final stasis vectors. All kernels all viable. $\overline{\cos} = 0.9854$ (9.8ř).

| | $k_0$ | $k_1$ | $k_2$ | $k_3$ |
|---|---|---|---|---|
| $k_0$ | — | 0.9948 | 0.9808 | 0.9607 |
| $k_1$ | — | — | 0.9955 | 0.9839 |
| $k_2$ | — | — | — | 0.9964 |
| $k_3$ | — | — | — | — |

Table 5: `s3_failure_resilience`: viable_all ratio before and after S3 freeze. 4 kernels, $\tau_0 = 0.80$, cgroup v2 active.

| **Phase** | **S3 active** | **viable_all** | **Temp range** |
|---|---|---|---|
| Phase 1 (0–3 s) | Yes | 100% (12/12) | 47–49°C |
| Phase 2 (3–6 s) | No | 100% (12/12) | 46–47°C |
| $\Delta$ | — | 0.0% | — |

Table 6: `critical_angle`: stasis_cos under uncontrolled thermal ramp. Hardware: i7-7600U, 4 workers, safety gate at 70°C.

| **Metric** | **Value** |
|---|---|
| Baseline temperature | 67.0°C |
| Temperature range | 65–69°C |
| viable_full throughout | **true** (never false) |
| Minimum stasis_cos observed | **0.9377** |
| Minimum angle (arccos) | **20.3°** |
| First nonviable tick | None (safety halt at 70°C) |
| $\theta_{\text{PSI}}$ bound | $< 0.9377$ |